Itô–Tanaka Certificates: Towards Scalable SDE Certification through Compositionality

Filip Morawiec · University of Birmingham Summer Research Programme · 2026

From exact reach–avoid guarantees to neural region discovery and local-time conditions by construction.

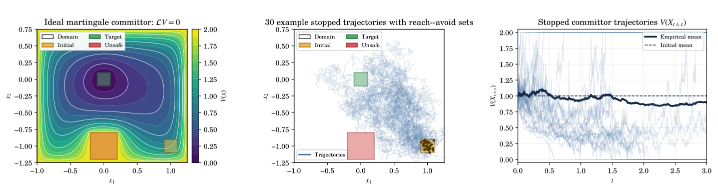
1. Introduction

Consider the problem of ensuring that a continuous system behaves safely under uncertainty—for example, that a humanoid robot does not fall over while walking. One way to ensure safety is to perform many experiments. For such complex systems, however, this is not feasible. Neural Continuous-Time Supermartingale Certificates [1] instead give exact probabilistic guarantees of safety for stochastic differential equations (SDEs) of the form

$$dX_t = f(X_t)\,dt + \sigma(X_t)\,dW_t, \qquad X_t \in K \subset \mathbb{R}^d,$$

over a compact domain $K \subset \mathbb{R}^d$.

An ideal certificate turns the stopped diffusion into a martingale



Problem setup

Given an initial set X_0, a target set X_T, an unsafe set X_u, and $\alpha < \beta$, assume we find a nonnegative function $V : K \to \mathbb{R}$ such that

$$\sup_{X_0} V \le \alpha < \beta, \qquad \inf_{X_u} V \ge \beta, \qquad \inf_{\partial K \setminus X_T} V \ge \beta.$$

Stop on reaching X_T, ∂K, or the superlevel set $\{V \ge \beta\}$.

Theorem (reach–avoid guarantee). If the stopped process $V(X_{t\wedge\tau})$ is a supermartingale, then

$$\mathbb{P}(\tau_{X_u} \wedge \tau_{\partial K} < \tau_{X_T}) \le \frac{\alpha}{\beta}.$$

Thus failure includes either reaching the unsafe set or leaving the certified domain before the target. Both imply reaching certificate level β by the stopping time.

2. Itô–Tanaka certificates

In practice, we want to find V using neural networks. To ensure that V is a supermartingale, we must check conditions relating V, f, and σ. Suppose V is continuous and twice differentiable inside polyhedral cells K_i. Expanding the multidimensional Itô–Tanaka formula separates the drift, martingale, and interface-local-time effects:

$$V(X_{t\wedge\tau}) = V(X_0) + \int_0^{t\wedge\tau} \mathcal{L}V(X_s)\,ds + M_t + \frac{1}{2}\sum_F \int_0^{t\wedge\tau} \gamma_F(X_s)\,dL_s^F,$$

where

$$\mathcal{L}V = \nabla V^\top f + \tfrac{1}{2}\operatorname{tr}(\sigma\sigma^\top \nabla^2 V), \quad \gamma_F = (\nabla V_j - \nabla V_i)^\top n_{i\to j,F}.$$

Theorem (Itô–Tanaka certificate). If the generator condition $\mathcal{L}V_i \le 0$ and the local-time condition $\gamma_F \le 0$ hold outside X_T and below β, then—under the standard integrability assumption—$V(X_{t\wedge\tau})$ is a supermartingale.

3. Piecewise-quadratic neural network

We parameterize the certificate as

$$V(x) = \sum_{k=1}^{m} \lambda_k \phi(z_k(x)) + c,$$

where

$$z(x) = \mathtt{NN}(x) = W_L\,\mathrm{ReLU}(\cdots \mathrm{ReLU}(W_1 x + b_1)\cdots) + b_L$$

and $\phi = \mathtt{PiecewiseQuadratic1D}$ is a fixed continuous piecewise-quadratic activation.

The hidden ReLU network partitions the input domain into polyhedral cells. On each cell, $z(x)$ is affine; after refining by the pieces of ϕ, the scalar certificate has the exact form

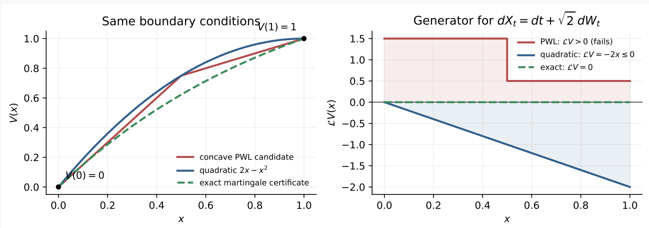
$$V_i(x) = c_i + p_i^\top x + \tfrac{1}{2}x^\top Q_i x, \qquad K_i = \{x : A_i x \le b_i\}.$$

Thus $\nabla V_i = p_i + Q_i x$ and $\nabla^2 V_i = Q_i$ are explicit, making the generator and interface conditions algebraic.

Why not a plain ReLU certificate? A plain ReLU network is only piecewise linear and has no curvature inside its cells. Consider $dX_t = dt + \sqrt{2}\,dW_t$ on $[0, 1]$, and require $V(0) < V(1)$. For continuous piecewise-linear V, the local-time condition makes the slopes nonincreasing. Yet increasing overall requires the first linear piece to have slope $a_i > 0$. Since $V'' = 0$ in its interior,

$$\mathcal{L}V = V' + \tfrac{1}{2}(2)V'' = a_i > 0,$$

contradicting the generator condition. Kinks alone cannot repair this failure.



Curvature repairs the failure. The quadratic $V(x) = 2x - x^2$ has the same endpoint values, while

$$\mathcal{L}V = (2 - 2x) + \tfrac{1}{2}(2)(-2) = -2x \le 0.$$

The negative curvature contribution compensates for the positive drift.

Core idea:

1. Checking the certificate conditions is much less expensive for ReLU networks.
2. Because we can handle a piecewise quadratic network, we can compose multiple certificates by taking their minimum.

4. Pairwise-interface bottleneck

Only adjacent full-dimensional regions sharing a facet contribute local time, but discovering N neural regions may already be exponential in depth and a naïve pairwise search costs $O(N^2)$. The baseline verifier constructs the actual adjacency graph, but the long-term goal is to avoid the facet check.

Guarantee the sign by construction

Split regular and singular curvature:

$$\boxed{V_\theta(x) = S_\theta(x) - C_\psi(x).}$$

S_θ is C^1 and piecewise quadratic, for example

$$S_\theta(x) = c + p^\top x + \tfrac{1}{2}x^\top Hx + \tfrac{1}{2}\sum_k v_k\,\mathrm{ReLU}(a_k^\top x + b_k)^2,$$

while C_ψ is convex and piecewise affine: either a ReLU input-convex neural network or $\max_r(a_r^\top x + b_r)$.

The C^1 branch has no singular surface curvature. Convexity gives

$$D^2_{\mathrm{sing}} V_\theta = -D^2_{\mathrm{sing}} C_\psi \preceq 0,$$

so every local-time jump has the required sign *without* discovering the facet adjacency graph. Value bounds and the interior generator still require verification.

Expressivity trade-off

The construction controls only the nonsmooth curvature: S_θ may retain indefinite ordinary curvature. However, global convexity of C_ψ is stronger than constraints only on reachable facets, and its CPWL gradient jumps are piecewise constant rather than affine along a general PWQ facet.

5. Current contribution

- Exact multidimensional neural-cell discovery with PWQ coefficients.
- A sound geometric 2D verifier for values, generator, continuity, and local-time jumps.
- Residual ICNN and max-affine architectures with the jump sign built in.
- A structural verifier that omits facet enumeration for these architectures.
- A successfully trained and exactly verified construction-safe certificate, shown below.

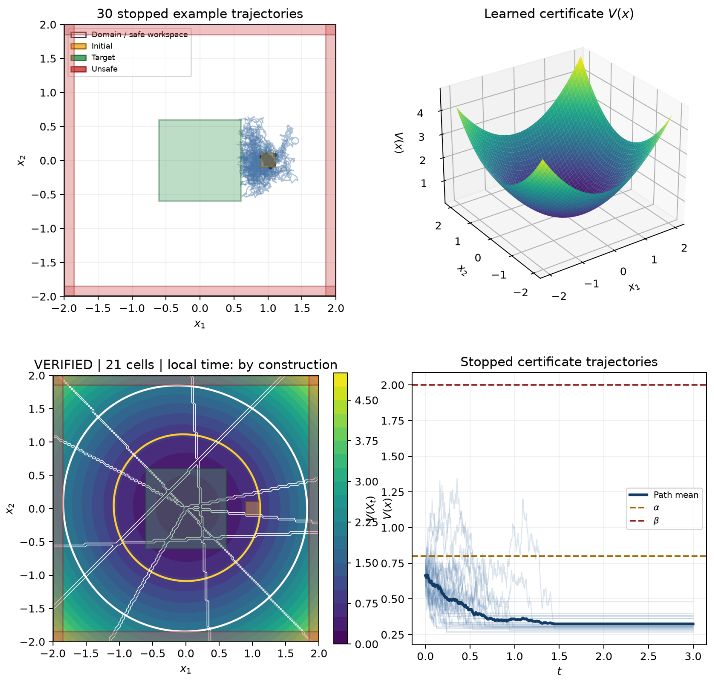
Exact verification currently targets 2D SDEs on hyperrectangular domains; scaling beyond explicit cell enumeration remains ongoing work.

6. Training a certificate

Controlled 2D OU benchmark

The experiment uses $dX_t = -X_t\,dt + 0.5\,dW_t$ on $K = [-2, 2]^2$.

General construction-safe certificate trained on the controlled OU benchmark



The four panels show: (1) marked reach–avoid sets and 30 stopped state trajectories; (2) the learned surface; (3) piecewise boundaries and 21 verified cells; and (4) the stopped processes $V(X_{t\wedge\tau})$. Simulations are diagnostic only.

● **Exactly verified.** The trained certificate has **21 PWQ cells**; all **4 max-affine pieces** are active, and the local-time condition holds by construction. With $\alpha = 0.8$ and $\beta = 2$, the certified failure-probability bound is $\alpha/\beta = 0.4$.

Training recipe

ADAM, LR 2×10^{-3}, BATCH 256, SEED 2026

$$\ell_0 = \max_{X_0}[V - (\alpha - m)]_+,$$
$$\ell_u = \max_{X_u}[(\beta + m) - V]_+,$$
$$\ell_{\partial K} = \max_{\partial K\setminus X_T}[(\beta + m) - V]_+,$$
$$\ell_+ = \max_K[-V]_+,$$
$$\ell_{\mathcal{L}} = \max_{K\setminus X_T} \min\{[\mathcal{L}V + \varepsilon + m_{\mathcal{L}}]_+, [(\beta + m) - V]_+\}.$$

$\ell = 80(\ell_0 + \ell_u + \ell_{\partial K} + \ell_{\mathcal{L}}) + 20\ell_+$ plus regularisation. Verifier counterexamples are added every 25 epochs; checkpoint 200 verifies exactly. No concavity loss is needed.

[1] G. Neustroev et al., "Neural Continuous-Time Supermartingale Certificates," AAAI 2025.

Contact: filip.morawiec@gmail.com · Middle-of-programme result; broader architectural comparisons remain in progress.